$$\begin{array}{r} 0{,}1587 \\ 0{,}8413 \\ \hline 1{,}0000 \end{array} \quad \begin{array}{r} 0{,}6915 \\ 0{,}3085 \\ \hline 1{,}0000 \end{array} \quad \begin{array}{r} 0{,}7734 \\ 0{,}5987 \\ \hline 1{,}3721 \end{array}$$

(10) 1. The annual salaries of employees in a large company are approximately normally distributed with a mean of \$50,000 and a standard deviation of \$20,000.

a) What percent of people earn less than \$40,000? (3 Points)

b) What percent of people earn between \$45,000 and \$65,000?

c) What percent of people earn more than \$70,000? (3 Points)

$$\begin{array}{r} 0{,}7734 \\ 0{,}5987 \\ \hline 1{,}3721 \end{array}$$

(a) $P(X < 40000) = P\left(\frac{X-50000}{20000} < \frac{-10000}{20000}\right) = P\left(Z < \frac{-1}{2}\right) = \Phi(-0{,}5)$

$= 1 - \Phi(0{,}5) = 1 - 0{,}6915 = 0{,}3085$

(b) $P(45000 < X < 65000) = P\left(\frac{-5000}{\underset{4}{\cancel{20000}}} < Z < \frac{\overset{2}{\cancel{15000}}}{\underset{4}{\cancel{20000}}}\right) = P(-0{,}25 < Z < 0{,}75)$

$= \Phi(0{,}75) - \Phi(-0{,}25) = \Phi(0{,}75) - (1 - \Phi(0{,}25))$

$= \Phi(0{,}75) + \Phi(0{,}25) - 1 = 0{,}7734 + 0{,}5987 - 1 = 0{,}3721$

(c) $P(X > 70000) = P\left(Z > \frac{20000}{20000}\right) = P(Z > 1) = 1 - P(Z \leq 1)$

$= 1 - \Phi(1) = 1 - 0{,}8413 = 0{,}1587$

(10) 2. The joint density function of X and Y is given by

$$f(x,y) = e^{-(x+y)} \qquad 0 < x < \infty,\ 0 < y < \infty$$
$$= 0 \qquad \text{otherwise}$$

Find the density function of the random variable X/Y.

$$P\left(\frac{X}{Y} \le a\right) = P(X \le aY) = \int_0^\infty \int_0^{ay} e^{-x-y}\,dx\,dy = \int_0^\infty e^{-y}\left(-e^{-x}\Big|_0^{ay}\right)$$

$$= \int_0^\infty e^{-y}\left(1-e^{-ay}\right)dy = \int_0^\infty e^{-y} - e^{-(1+a)y}\,dy$$

$$= -e^{-y}\Big|_0^\infty - \frac{-1}{(a+1)}\,e^{-(a+1)y}\Big|_0^\infty = 1 - \frac{-1}{a+1}\cdot(-1) = 1 - \frac{1}{a+1} = \frac{a}{a+1}$$

$$f_{\frac{X}{Y}}(a) = \frac{d}{da}\,\frac{a}{a+1} = \frac{(a+1)-a}{(a+1)^2} = \frac{1}{(a+1)^2}$$

(10) 3. A man and a woman decide to meet at a certain location about 12:30 P.M. If the man arrives at a time uniformly distributed between 12:15 and 12:45 and if the woman independently arrives at a time uniformly distributed between 12:00 and 1 P.M., find the probability (a) that the first to arrive has to wait no longer than 15 minutes. (b) What is the probability that the man arrives first?

(a)

$|X-Y| \le 15 \rightarrow -15 \le X-Y \le 15 \rightarrow -15+Y \le X \le 15+Y$

$f_{(x,y)} = f_x \cdot f_y = \frac{1}{30} \cdot \frac{1}{60} = \frac{1}{1800}$

$\int_{15}^{45} \int_{y-15}^{y+15} \frac{1}{1800} \, dx \, dy = \int_{15}^{45} \frac{\cancel{30}^{1}}{\cancel{1800}_{60}} \, dy = \frac{1}{60} \cdot 30 = \frac{1}{2}$ #

(b)

$P(X<Y) = \int_{15}^{45} \int_{X}^{60} \frac{1}{1800} \, dy \, dx = \int_{15}^{45} \frac{60-X}{1800} \, dx$

$= \frac{X}{30} \Big|_{15}^{45} - \frac{X^2}{3600} \Big|_{15}^{45} = \frac{30}{30} - \frac{\cancel{60} \cdot 30}{\cancel{3600}_{60}}$

$= 1 - \frac{1}{2} = \frac{1}{2}$ #

$\int_t^1 1\,dx = (1-t) \quad \int_0^t 1\,dx = t$

(10) 4. Let $X_1, X_2, X_3, \ldots X_n$ be independent random variables uniformly distributed on (0,1). (A) Determine the probabiity of distribution (Cumulative Distribution Function) of Min ($X_1, X_2, \ldots X_n$) (B) Determine the prbability distribution (Cumulative Distribution Funct of Max($X_1, X_2, \ldots X_n$)

(A)

$P(\mathrm{Min}(X_1, \ldots X_n) \le t)$

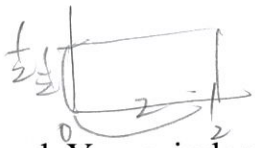
$= 1 - p(\mathrm{Min}\{X_1, \ldots X_n\} > t) = 1 - p(X_1 > t)\, p(X_2 > t) \cdots p(X_n$

$= 1 - (1-t)^n$ ※

(B) $p(\mathrm{Max}(X_1, \ldots X_n) \le t)$

$= p(X_1 \le t)\, p(X_2 \le t) \cdots p(X_n \le t)$

$= t^n$ ※



$f_X(a-y) = F(X+Y \le a)' = F(X \le a-y)$

$\int_0^{a-y} \frac{1}{2}\, dx = \frac{a-y}{2}$

(10) 5. If X and Y are independent random variables, both uniformly distributed on (0,2), calculate the probability density of X + Y.

$0 < X+Y < 4$

$\frac{1}{2}$

$f_{X+Y}(a) = \int_0^2 f_X(a-y) f_Y\, dy = \int_0^2 f_X(a-y) \frac{1}{2}\, dy$

$f_X(a-y) = F'(X+Y \le a) = F'(X \le a-y) = \left(\int_0^{a-y} \frac{1}{2}\, dx\right)' = \left(\frac{a-y}{2}\right)' = \frac{1}{2}$

当 $0 \le a \le 2 = f_{X+Y}(a) = \int_0^a \frac{1}{4}\, dy = \frac{a}{4}$

当 $2 < a < 4 = f_{X+Y}(a) = \int_{a-2}^2 \frac{1}{4}\, dy = \frac{2-(a-2)}{4} = \frac{4-a}{4}$

其它 $= f_{X+Y}(a) = 0$

(10) 6. . A system with three independent components works correctly if at least one component is functioning properly. Failures rates of the individual components are $\lambda_1 = 0.01$, $\lambda_2 = 0.02$ and $\lambda_3 = 0.04$ (assume exponential lifetime distributions)

(a) Determine the probability that the system will work for 100 hours.

(b) Determine the density function of the lifetime X of the system.

$\int_0^t \lambda e^{-\lambda x} dx$

$= \lambda \cdot \frac{1}{-\lambda} e^{-\lambda x} \Big|_0^t$

$= -(e^{-\lambda t} - 1)$

$= 1 - e^{-\lambda t}$

(a) $p(X \geq 100) = 1 - p(X < 100) = 1 - (1-e^{-x})(1-e^{-2x})(1-e^{-4x})$

$1 - (1-e^{-1})(1-e^{-2})(1-e^{-4})$

(b) p(system still work at x)

$= 1 - p($ system fails at x $)$

$= 1 - (1-e^{-0.01x})(1-e^{-0.02x})(1-e^{-0.04x})$

(10) 7. The joint density function of X and Y is given by

$$f(x,y) = 2\,e^{-x}\,e^{-2y} \qquad 0 < x < \infty,\ 0 < y < \infty$$
$$= 0 \qquad \text{otherwise}$$

Compute (a) $P\{X > 1, Y < 1\}$, (b) $P\{X < Y\}$, and (c) $P\{X < a\}$

(a) $p(x>1, y<1) = \int_0^1 \int_1^\infty 2e^{-x} e^{-2y}\,dx\,dy = \int_0^1 2e^{-2y}\left(-e^{-x}\Big|_1^\infty\right)dy$

$= \int_0^1 2e^{-2y} \cdot e^{-1} \cdot dy = 2e^{-1} \cdot \frac{1}{-2} e^{-2y}\Big|_0^1 = -e^{-1}(e^{-2}-1) = e^{-1} - e^{-3}$ #

(b) $p(x<y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy = \int_0^\infty 2e^{-2y}\left(-e^{-x}\Big|_0^y\right)dy$

$= \int_0^\infty 2e^{-2y}(1-e^{-y})\,dy = \int_0^\infty 2e^{-2y} - 2e^{-3y}\,dy$

$= -e^{-2y}\Big|_0^\infty + \frac{2}{3}e^{-3y}\Big|_0^\infty = 1 + \frac{2}{3}(0-1) = 1 - \frac{2}{3} = \frac{1}{3}$ #

(c) $p(x<a) = \int_0^a \int_0^\infty 2e^{-x}e^{-2y}\,dy\,dx$

$= \int_0^a 2e^{-x}\left(\frac{1}{-2}e^{-2y}\Big|_0^\infty\right)dx$

$= \int_0^a 2e^{-x}\left(0+\frac{1}{2}\right)dx = \int_0^a e^{-x}\,dx = -e^{-x}\Big|_0^a$

$= 1 - e^{-a}$ #

$$\begin{array}{r} 0,1587 \\ 0,8413 \\ \hline 1,0000 \end{array}$$

(10) 8. (The Central Limit Theorem) An instructor has 50 exams that will be graded in sequence. The times required to grade the 50 exams are indendent, with a common distribution that has mean 20 minutes and standard deviation 5 minutes. Approximate the probability that the instructor will grade at least 25 of the exams in the the first 475 minutes of work.

$E(X) = 20 \times 25 = 500$, $\sigma = \sqrt{25 \times 25} = 25$

$$p(X_1 + \cdots + X_{25} \leq 475) = p\left(\sum_{i=1}^{25} X_i \leq 475\right) = p\left(\frac{\sum_{i=1}^{25} X_i - 500}{25} \leq \frac{-25}{25}\right)$$

$$= p(Z \leq -1) = \Phi(-1) = 1 - \Phi(1) = 1 - 0,8413$$

$$= 0,1587$$

※

(10) 9. If X and Y are independent Poisson random variable with respective parameters λ_1 and λ_2, compute the probability mass function of X + Y.

$$P(X+Y=n) = \sum_{i=0}^{n} P(X=i)P(Y=n-i) = \sum_{i=0}^{n} e^{-\lambda_1} \cdot \frac{\lambda_1^{i}}{i!} \cdot e^{-\lambda_2} \cdot \frac{\lambda_2^{(n-i)}}{(n-i)!}$$

$$= e^{-(\lambda_1+\lambda_2)} \cdot \sum_{i=0}^{n} \frac{n!}{i!(n-i)!} \lambda_1^{i} \lambda_2^{n-i} \cdot \frac{1}{n!}$$

$$= e^{-(\lambda_1+\lambda_2)} \cdot \frac{(\lambda_1+\lambda_2)^n}{n!}$$

※

(10) 10. Two fair dice are rolled. Find the joint probability mass function of X and Y when

(a) X is the largest value obtained on any die and Y is the sum of the values (5 Points)

(b) X is the value on the first die and Y is the larger of the two values (5 Points)

(a) $X=i,\ Y=j$

$i \backslash j$	2	3	4	5	6	7	8	9	10	11	12
1	$\frac{1}{36}$	0	0	0	0	0	0	0	0	0	0
2	0	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0	0	0
3	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0	0	0
4	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0	0	0
5	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	0	0
6	0	0	0	0	0	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(b) $X=i,\ Y=j$

$i \backslash j$	1	2	3	4	5	6
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
2	0	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
3	0	0	$\frac{3}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
4	0	0	0	$\frac{4}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
5	0	0	0	0	$\frac{5}{36}$	$\frac{1}{36}$
6	0	0	0	0	0	$\frac{6}{36}$

心想事成 As You Wish

TABLE 5.1: AREA $\Phi(x)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF X

X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998